

DR JOSEPH MANGHAN

SENIOR FRONTEND ENGINEER | CONTEXT ENGINEERING

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Senior Frontend Engineer specialising in context engineering and structured prompt-driven development. Experienced in building AI-first workflows that improve team productivity and code quality. Passionate about UX, responsive design, and iterative improvement. Expertise across the full SDLC.

Experience

Senior Frontend Engineer

Backbase / Jan 2026 - present

Context engineer at Backbase — authoring its structured prompt-driven development framework, training across the context engineering space, alongside senior frontend delivery in the Identity Access Management space.

The framework — a playbook of agent skills carrying a feature from ticket refinement to vetted PR — rolled out to hundreds of engineers across the company:

- Playbook follows QRSPI principles — compacting between disambiguation, codebase research, planning, and implementation — with harness feedback built in.
- Build coding agent utilities (compound engineering, issue generation, value capture handoffs) and a subagent layer for focused tasking, model selection and cost-targeted dispatch.
- Deliver insights and patterns on outer harness practice — feedforward, feedback, and back-pressure.
- Develop prompt evaluation methods that test effectiveness, benchmark across models, and catch drift between iterations.
- Iterate on the framework as the field matures — researching new patterns, tuning skills, aligning with best practices.

Teach context engineering across the company — workshops, written documentation, and per-team adoption support.

In the IAM space:

Led the technical design for complex IAM features, orchestrating input from UX, BE and Tech Writers.

Oversaw the organisation, story mapping and UX liaison of a major refactor initiative using a shared componentisation strategy:

- Significantly improved code quality and UX parity across the product

Frontend Engineer

Backbase | Jan 2023 - Dec 2025

Consolidated frontend expertise through hands-on delivery, quality advocacy, and iterative improvement in IAM feature development.

Managed prompt assets, context, and specs at squad level to support day-to-day feature delivery.

Conducted an a11y audit to identify gaps against the WCAG AA standard:

- Tools included WAVE, Playwright E2E accessibility packages, and manual screen reader reviews

Proposed and delivered an operational tech debt initiative that improved our capacity to iterate on features

Junior Frontend Engineer

Backbase | Aug 2021 - Dec 2022

Web Development Internship

Rhys Welsh Web Design | Jun - Jul 2021

Foundational experience with CMS content architecture.

PhD in Web Science & Music

University of Southampton | 2015 - 2020

Utilised web practices to outsource compositional tasks to crowds.

 Skills	 Education
Coding Agents: Context Engineering • Harness Engineering • Structured Prompt-driven Development • Compound Engineering • Prompt Evaluation • Agent Skills • Cursor • Claude Code • Gems	University of Southampton PhD in Web Science & Music 2015-20
Technical: Typescript • Angular • React • Jest • Vite • Playwright • Cypress • SCSS	Pitman Training The Tech Academy Web Developer Diploma **(Dist.)** 2020-21
Professional: Strong written & verbal communication • Mentorship • Public speaking • Research & analysis	MMus Music Composition (Dist.) 2014-15 BA Music (1st class) 2011-14